
Research on green supply chain decision-making considering government subsidies and blockchain technology

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Abstract

Under the dual-carbon framework, the concept of green consumption has become deeply embedded in people's mind, and the world is increasingly emphasizing the development of green supply chains. However, consumers often face uncertainty regarding the environmental friendliness of products. Blockchain technology, with its capabilities for tracking, traceability, and information transparency, can effectively address this issue. It promotes the integration of "dual chains" to optimize resource allocation. Within the context of the green supply chain system comprising a manufacturer and a retailer, this paper investigates the uncertainty of consumers' valuation of green products, the heterogeneous preference for blockchain technology and other factors, constructing a Stackelberg game model to examine the impacts of blockchain technology and government subsidies on the decision-making within a green supply chain. Firstly, the optimal decision-making challenges faced by the traditional supply chain without government subsidies and the blockchain technology-supported supply chain are investigated. This is followed by a comprehensive analysis of the optimal decision-making for the green supply chain under two scenarios: the government's technology subsidies and output subsidies. The result shows that (1) in the scenario without government subsidies, manufacturers will adopt blockchain technology only if the operating cost of the blockchain is below a certain threshold. This threshold is influenced by consumers' heterogeneous preferences for blockchain technology, consumer uncertainty, as well as some other factors. When manufacturers adopt blockchain technology, demand does not always increase. Specifically, demand is related to the operating cost of the blockchain; demand increases when the operating cost is lower, and conversely, it decreases when the operating cost is higher. (2) When the government provides subsidies for technology or output, manufacturers are more inclined to adopt blockchain technology if its operating costs are below a certain threshold. Moreover, these thresholds are directly proportional to the value of the government subsidy coefficient, indicating that government subsidies can encourage the adoption of blockchain technology in the supply chain. Therefore, enterprises should take into account factors such as government subsidies and consumer preferences for blockchain technology when deciding on its adoption.

Keywords: Green Supply Chain ;Government Subsidies ;Blockchain

1. Introduction

As China's economy and society advance towards high-quality development, the challenge of achieving a win-win situation in terms of economic and environmental benefits has become urgent. The green supply chain, emblematic of Chinese modernization that promotes harmony between humans and nature, has ascended to the level of a national strategy. The Central Economic Work Conference pointed out that: accelerate the creation of a green and low-carbon supply chain, build a new type of energy system, and continue to fight a good battle to defend the blue sky, blue water and clean soil. With the gradual deepening of the concept of green consumption, consumers usually consider greenness when purchasing products, and are still more willing to buy green products even when the price is high (Hong et al., 2019). With the government's heightened focus on environmental issues and consumers' growing awareness of the environment, an increasing number of manufacturers are producing green products and striving to clearly differentiate them from traditional products. However, due to information asymmetry, consumers may doubt the environmental credentials of these green products, leading to uncertainty. This uncertainty can diminish consumers' willingness to purchase and, consequently, impact the demand for green products.

Blockchain technology offers a promising solution to the issues mentioned above. In recent years, blockchain has found applications across various sectors. As a decentralized, secure, and traceable distributed ledger technology, it can deliver transparent, accurate, and reliable product information to both enterprises and consumers. Its unique benefits present new opportunities for the development of green supply chains. For example, IKEA designs and produces a table whose raw materials come from sustainable forests in Indonesia. By using blockchain technology, IKEA can ensure that the entire process is controlled and that the final product is indeed made from the specific wood described (Saberli et al., 2019). More and more companies are increasing consumer trust by applying blockchain technology to trace product information, increase transparency of green products, and reduce or eliminate uncertainty in their valuation of products. However, not all companies opt to implement blockchain technology in practice, owing to the additional costs associated with its adoption and the varying preferences of consumers towards its use. Therefore, it's essential to investigate whether manufacturers decide to adopt blockchain technology in scenarios where there is consumer uncertainty about green products and differing preferences for the application of blockchain technology.

The government can encourage green production and improve social welfare by providing moderate low-carbon subsidies to enterprises. For example, in 2021, Hunan Province will provide a one-time bonus of 500,000 RMB to enterprises that have obtained national-level green factories, green products, green parks and green supply chain management. The government's low-carbon subsidies to enterprises are mainly in the form of green technology input subsidies (hereinafter referred to as

technology subsidies) and green product output subsidies (hereinafter referred to as output subsidies). The former is a behavioral subsidy that encourages more enterprises to invest in green technology R&D; while the latter is a result-based subsidy that provides incentives to enterprises that have successfully developed green products. Government subsidies are also one of the important means to promote the development of green supply chains. In green supply chains, different subsidy modes may affect the pricing and product greenness strategies of the supply chain, which in turn may affect the conditions for the adoption of blockchain technology, i.e., the government's attitude towards the introduction of blockchain technology by enterprises in the green supply chain is particularly important. Therefore, it is necessary to examine the impact of blockchain technology and government subsidies on green supply chains, considering the combined effects of both blockchain technology and government subsidies.

To address the issues mentioned above, our approach involves two main steps. First, we explore the optimal strategies for retail price, wholesale price, and greenness input level of green products in both traditional and blockchain-supported supply chains without government subsidies. This consideration includes factors such as consumer uncertainty about the product value and the operating cost of blockchain technology. Second, we incorporate variables like the degree of consumer preference heterogeneity for blockchain technology and examine the optimal strategies for relevant stakeholders in traditional and blockchain-supported supply chains under scenarios involving government technological and production subsidies, followed by a comprehensive analysis. The primary contributions of this paper are outlined as follows:

(1) Without government subsidies, manufacturers opt for blockchain technology when its operating costs are low. However, the adoption of blockchain technology does not always lead to higher demand.

(2) With government technology subsidies, manufacturers are inclined to adopt blockchain technology only if its operating cost falls below a certain threshold. This threshold increases monotonically with the degree of consumer preference for blockchain technology and the coefficient of the government's technology subsidy.

(3) When the government provides subsidies for manufacturers' output, manufacturers again will adopt blockchain technology only if its operating cost is below the threshold. This threshold is influenced by the degree of consumer preference for blockchain technology, the government's technology subsidy coefficient, consumer uncertainty, and it increases monotonically with the government's output subsidy coefficient.

The structure of the paper is as follows: Section 2 offers a review of related literature. Section 3 details the problem description and formulates the hypotheses. Section 4 analyzes the equilibrium outcomes for both traditional and blockchain-enabled supply chain models in scenarios without

government subsidies. Section 5 delves into the equilibrium outcomes of the model under government subsidies, distinguishing between technology subsidies and output subsidies. Section 6 provides a summary of the entire paper.

2. Literature review

This paper reviews literature in three main areas: studies on green supply chains, research on the application of blockchain technology in supply chains, and analyses of the impact of government subsidies on supply chains.

With regard to green supply chains, some scholars have examined the progress and impacts of green supply chains on macroeconomic management and sustainable development. Li (2024) modeled the complex relationship between environmental sustainability and economic development and found that firms adopting a global sustainable consumption and production framework showed greater consistency with China's sustainability goals and were able to change as the fundamental conflict between the need for economic growth and the need for environmental sustainability changed, while China's complex supply chain management of finance highlighted the difficulties faced by firms. The fundamental conflict between the need for economic growth and the need for environmental sustainability, and the complexity of managing finance in China's supply chains highlight the difficulties faced by firms, and the study ultimately provides a comprehensive understanding of the critical balancing act of China's path towards environmental sustainability. It emphasizes the importance of green supply chain management. Raghu et al. (2023) created SDG citation network by mapping publications from 2013 to 2022 to 17 SDGs using SDG mapping algorithm. The results of the study show that green supply chain is mainly focused on three areas of research: industry 4.0 and sustainable supply chain, green supply chain time coordination, and sustainable environmental policy management. From the study, it can be seen that international interest and cooperation in green supply chain research covers Europe, Western countries, the Middle East, South America and Africa. Rahman et al. (2023) discusses the impact of in-house environmental management, green procurement, eco-design and collaboration with suppliers and customers of green supply chain management on the sustainable performance of firms. Data was collected from 190 respondents from manufacturing firms located in Khyber Pakhtunkhwa province of Pakistan by using structured questionnaire. The study found that in-house environmental management, green procurement and eco-design have a significant positive impact on firms' sustainable performance, while the positive impact of cooperation with suppliers and customers on firms' sustainable performance is not significant. Bhatia and Gangwani (2020) provides an overview of empirical research on green supply chain management. The results indicate that there are still several aspects of empirical research that remain unexplored, such as the use of actual data for green supply chain management performance analysis, the use of longitudinal data,

and the use of research designs other than surveys, and that sectors other than manufacturing and automotive are still relatively unexplored. The collection of data for samples from emerging economies also needs more attention.

Some scholars have also studied the progress and impact of green supply chain on business economy. Ying et al. (2024) explores a specific government-led green supply chain demonstration project focusing on the financial impacts of such initiatives, and through a double difference model and related data, the results indicate a significant reduction in firms' financial distress, a result that explains how the value of green policies permeates within firms, and in addition the study highlights the empirical link between sustainable operations and financial resilience, providing a theoretical basis for firms to actively adopting green practices and assisting the government in formulating green policies by providing a theoretical basis. Wu et al. (2023) constructed a joint product pricing-ordering-greenness decision-making problem and considered the effect of demand uncertainty, provided an analytical expression for the optimal solution to the stochastic model, and proposed a bargaining game on wholesale price among supply chain members to coordinate the decision-making. It is found that although demand uncertainty leads to inefficiency in green supply chains, it may have a positive impact on product greenness and price. Halat et al. (2023) has implemented various climate change policies for the industrial sector in response to the duty given to the government to study the pricing and ordering competition between two green supply chains taking into account the carbon emission rules. The article develops a mathematical model giving the structure of the supply chain and the carbon emission policies, providing a series of insights highlighting the impacts of the carbon emission regulations on the prices, the order quantities, the transportation lead time, the carbon emissions, and the welfare of the society. Mondal and Giri (2024) attempted to combine product complementarity and bundling strategies with green innovation and remanufacturing in a two-tier supply chain. Based on the manufacturer-dominated Stackelberg game-theoretic approach, two models were constructed considering the retailer's bundling strategy, and the optimal responses and profitability were analyzed and numerically compared under various models, and the results showed that in the Stackelberg scenario, the decision-making of the two manufacturers will depend more on product complementarity. The results show that in the Stackelberg scenario, the decisions of the two manufacturers will depend more on product complementarities, and secondly, regardless of the bundling strategy, the sequential decision-making scheme favors the dominant manufacturer, while the joint decision-making scheme brings profits to the other manufacturer, the retailer and the whole supply chain. Most of the literature highlights the importance of green supply chain management in macroeconomic management and sustainable development, and has received global focus, so this paper chooses to analyze the research based on green supply chain.

Blockchain technology has been widely used in the supply chain. Liu et al. (2021) describe the

impact of blockchain technology on enterprises through a dynamic game model based on the traceability of blockchain and analyze the choice of sales model of e-commerce platforms under channel competition; Xu et al. (2023) discusses the coordination strategy under dual-channel of the platform supply chain after considering the impact of blockchain technology on green investment; while Xu and Duan (2022) constructed a green supply chain game model based on blockchain technology to address the problem of consumers' incomplete trust in green products, and revealed the sufficient conditions for blockchain implementation by comparing the two scenarios of non-implementation and implementation of a green supply chain. Balan et al. (2021) explored the demand for blockchain in supply chain management under the environment of Industry 4.0 from the perspective of big data, and found that the blockchain implementation can grow revenue and reduce carbon emissions. It also identifies key implementation guidelines and issues for blockchain in supply chain management, which can help to drive the body of knowledge towards big data and blockchain. Piera et al (2022) propose an integrated three-tier revisit framework for designing circular blockchain platforms, bridging the three inverse processes of the circular supply chain and the three factors influencing blockchain technology, and emphasize blockchain as a tool for improving waste stream control and technological capabilities for product return management activities. Bai et al. (2021) developed a green data supply chain based on blockchain using a game theory approach with trust management to document sensor trust. The results show that blockchain can maintain the proper operation of the system more securely and manage a large number of devices efficiently. Cui et al. (2023) analyze different ways of using blockchain to improve supply chain transparency and explore the conditions for supply chains to adopt blockchain to promote cooperative competition and shared information decision making with others. Xu et al. (2023) use game theory methods to analyze manufacturers' and platforms' s optimal pricing decisions and strategies to explore the impact of consumer preference for overseas green products and supply chain performance with the implementation of blockchain technology. Wu et al. (2023) explore the impact of the application of blockchain technology on different financing strategies of manufacturers by constructing a green supply chain model. Song and Yan.(2023) analyze the optimal operation decisions of green supply chains of two manufacturers under different competitive and cooperative relationships by considering the impact of government subsidies and green trust. Most of the above studies analyze the role of blockchain technology and its impact on the supply chain or green supply chain decision-making, but few papers analyze it in conjunction with the factor of government subsidies, and almost no articles consider the heterogeneous preferences of consumers for blockchain technology applications. In this paper, we will analyze the decision-making problem of green supply chain and the optimal conditions for the application of blockchain technology by considering the two types of government subsidies and introducing the factors of consumers' preference for blockchain technology.

In recent years, scholars have conducted extensive research on government subsidies in supply chains. Huang et al. (2020) study three kinds of government subsidy policies, namely, green credit, production subsidy and sales subsidy, in a financially constrained green supply chain, and elucidate the optimal conditions of different subsidy modes. Zhang et al. (2023) study the impact of government subsidies on the decision-making of a green supply chain consisting of a manufacturer and a (2023) investigates the impact of government subsidies on the decision-making of a green supply chain consisting of a manufacturer and a retailer, including input subsidies and output subsidies. Zheng et al. (2023) analyzes the optimal subsidy modes of the government and enterprises, including product subsidies, innovation subsidies, and product innovation subsidies, and explores the impacts of different subsidy modes on the government subsidy strategy and the enterprise's pricing strategy. Shi et al. (2024) based on the farmer's self-invested green technology and outsourced green technology two green production strategy selection scenarios, and study the impact of government subsidies on two-tier agricultural supply chains, where the government provides two types of subsidies: green technology operation subsidies and green technology investment subsidies. Abhijit et al. (2023) used the Stackelberg game approach to evaluate centralized and decentralized marketing strategies with the (2023) used the Stackelberg game method to evaluate centralized and decentralized marketing strategies and analyzed the optimal pricing strategy to maximize the overall profit of the green dual-channel supply chain in the presence and absence of government subsidies. Guo et al. (2023) constructed a two-tier e-commerce supply chain with one farmer and one e-commerce platform, and examined the effects of five different government subsidies on the greenness of agricultural products, the wholesale price, the selling price, and the profitability of the supply chain based on the Stackelberg game theory. profit. Song et al. (2023) analyzed the impact of manufacturers' equity concerns on manufacturers' and retailers' decision-making in a dual-channel green supply chain under government subsidies by constructing a retailer-driven Stackelberg game model considering the existence of manufacturers' equity concerns. Li et al. (2022) considered that in the presence of differences in sales strategies and government subsidies Li et al. (2022) investigated the operational efficiency of a platform supply chain by constructing four game models in which the manufacturer's green production and the platform's marketing activities are carried out simultaneously under different sales strategies and government subsidies. Sha and Liu (2022) investigated the decision-making of a price and green innovation game in a supply chain of complementary products and the optimization of the coordination of the supply chain in the case of green subsidies from the government to the retailers based on the game theory. 2022) used Stackelberg game theory and numerical simulation analysis to study the impact of government subsidies and altruistic preferences on green supply chain innovation, and analyzed the optimal decisions of enterprises under different decision models. Government subsidies play an important role in supply chains, and most of the literature analyzes the different subsidy

methods and targets of government subsidies, but few scholars combine blockchain technology to study the decision-making problem of green supply chains. In this paper, we will consider the optimal decision-making of traditional supply chains and blockchain technology-supported supply chains with or without government subsidies.

In summary, there is a large body of literature on decision making in green supply chains under the influence of different factors, and the application of blockchain technology in supply chains has also been a popular area of research in recent years, but there is relatively little literature on the application of blockchain technology in green supply chains considering the presence or absence of government subsidies, and we also consider the heterogeneous preferences of consumers for the application of blockchain technology, the operation of blockchain technology cost, and other factors.

3. Model description

In this paper, we consider a secondary green supply chain consisting of a manufacturer M , and a retailer R . The manufacturer produces green products and distributes them to the retailer at a unit wholesale price w and the retailer then sells them to consumers at a price p . In addition blockchain technology is widely used in the green supply chain, and consumers can trace their products through blockchain technology to enhance the level of green trust. Manufacturers decide whether to adopt blockchain technology or not, so this paper assumes the existence of two green supply chains, namely, the traditional supply chain T (the supply chain without blockchain technology) and the blockchain-supported supply chain B . Based on the description of the consumer utility function in references Xu and Duan (2023), we, assume that the consumers' assessment of the base value of the product v is heterogeneous and obeys a uniform distribution between 0 and 1. In the traditional supply chain, the consumer utility function can be expressed as:

$$U^T = v - p^T + (1 - \lambda)g^T \quad (1)$$

Due to the limited information and the opacity of green products, consumers are often unsure of the value of green products. based on previous studies (Wu et al., 2021; Niu et al., 2021), without the introduction of blockchain technology, we use $\lambda(0 \leq \lambda < 1)$ to denote the degree of consumer uncertainty in Equation (1), and the greater the the greenness of the product. g denotes the greenness of the product. When a manufacturer chooses to produce a green product, each unit of greenness requires the manufacturer to invest a green R&D cost of $kg^2/2$, $k(k > 0)$ is the cost coefficient of green R&D. In addition, it is assumed that the manufacturer's unit production cost is c .

When the supply chain adopts blockchain technology, utilizing its traceability mechanism and transparency, consumers can obtain full information about green products, when $\lambda = 0$. Referring to Tao et al (2023),for the description of the consumer utility function after applying blockchain, when the

enterprise adopts blockchain technology for green products, the consumer purchases the product: the utility function is:

$$U^B = \theta v - p^B + g^B \quad (2)$$

where $\theta(0 \leq \theta < 1)$ denotes the degree of consumer preference for the application of blockchain technology, and the larger θ is, the higher the degree of consumer preference for blockchain is, and the more willing they are to purchase blockchain-enabled green products. In addition, we assume that the cost of adopting blockchain is its operating cost, denoted by b as the unit operating cost of blockchain, which is mainly used for storing and updating data by creating and inputting blockchain and borne by the manufacturer.

The relevant parameters of this study are shown in Table 1.

Table 1

Notations.

Category	Notation	Description
parameter	v	The basic value of the product, $v \sim U[0,1]$
	λ	The degree of consumer uncertainty about the product value, $0 \leq \lambda < 1$
	D	Market demand
	θ	Degree of preference for the application of blockchain technology, $0 \leq \theta < 1$
	k	The cost coefficient of green production, $k \geq 0$
	b	Unit operational cost of blockchain, $0 \leq b < 1$
	c	The manufacturer's unit production cost
	μ	Amount of subsidy for individual low-carbon products
variable	τ	Green technology input subsidy coefficient, $0 \leq \tau < 1$
	w	Manufacturer's wholesale price per unit of green product
	g	The product's greenness level
	p	Retail price at the retailer
Other symbol	Π	Profit of each party involved
	subscript M	Manufacturer
	subscript R	Retailer
	superscript T	Traditional supply chain
	superscript B	Blockchain-supported supply chain
	superscript I	Technology subsidy situation
	superscript J	Output subsidy situation

4. Equilibrium analysis of green supply chain without government subsidy

The main research in this chapter is to analyze the equilibrium decision of traditional supply chain and the equilibrium decision of supply chain with blockchain technology in the absence of government subsidy..

4.1. The traditional supply chain model

When $U^T \geq 0$, i.e. $v \geq p^T - (1 - \lambda)g^T$, the consumer will choose to buy the green products, at which time the demand is $D^T = P\{U^T \geq 0\}$, thus obtaining the demand function:

$$D^T = 1 - p^T + (1 - \lambda)g^T \quad (3)$$

The retailer's profit function:

$$\Pi_R^T = (p^T - w^T)D^T \quad (4)$$

The manufacturer's profit function:

$$\Pi_M^T = (w^T - c)D^T - \frac{1}{2}kg^{T^2} \quad (5)$$

For many green products, the manufacturer dominates the supply chain and the retailer becomes a follower in the supply chain due to the fact that the manufacturer owns the primary key technology for green production, which gives them strong bargaining power over the retailer, and the two constitute a Stackelberg game.

Lemma 1. In a traditional supply chain, first solve the response function of the retailer (as a follower) given w_T and g_T . By $\frac{\partial \Pi_R^T}{\partial p_T} = 0$, we have $p_T^*(g_T, w_T) = -\frac{1}{2}g_T\lambda + \frac{1}{2}g_T + \frac{1}{2}w_T + \frac{1}{2}$. By substituting equation $p_T^*(g_T, w_T)$ into equation (4), the Hessian matrix of Π_M^T with respect to w_T and g_T ,

$$H(\Pi_M^T) = \begin{bmatrix} -\frac{\lambda}{2} + \frac{1}{2} & -1 \\ -k & -\frac{\lambda}{2} + \frac{1}{2} \end{bmatrix}. \text{ We can deduce that the Hessian matrix is negative definite when}$$

$k > \left(-\frac{\lambda}{2} + \frac{1}{2}\right)^2$. Therefore, when $k > \left(-\frac{\lambda}{2} + \frac{1}{2}\right)^2$, Π_M^T is jointly concave in w_T and g_T , can be obtained

the equilibrium results are as follows: (1) Retail price, greenness level and wholesale price: $p^{T*} =$

$$\frac{3k - c\lambda^2 + 2c\lambda + (k-1)c}{-\lambda^2 + 4k + 2\lambda - 1}, \quad g^{T*} = \frac{(c-1)(-\lambda+1)}{-\lambda^2 + 4k + 2\lambda - 1}, \quad w^{T*} = \frac{2k - c\lambda^2 + 2c\lambda + (2k-1)c}{-\lambda^2 + 4k + 2\lambda - 1}. \quad (2) \text{ Demand: } D^{T*} = \frac{k(1-c)}{-\lambda^2 + 4k + 2\lambda - 1}. \quad (3)$$

$$\text{Profits: } \Pi_R^{T*} = \frac{k^2(c-1)^2}{(-\lambda^2 + 4k + 2\lambda - 1)^2}, \quad \Pi_M^{T*} = \frac{k(c-1)^2}{-2\lambda^2 + 8k + 4\lambda - 2}.$$

Proposition 1. It can be clearly found that the equilibrium solution to the first order derivative of a has the following result: $\frac{\partial p^{T*}}{\partial \lambda} < 0$, $\frac{\partial w^{T*}}{\partial \lambda} < 0$, $\frac{\partial g^{T*}}{\partial \lambda} < 0$, $\frac{\partial \Pi_R^{T*}}{\partial \lambda} < 0$, $\frac{\partial \Pi_M^{T*}}{\partial \lambda} < 0$.

Proposition 1 deduce that p^{T*} , w^{T*} , g^{T*} , Π_R^{T*} and Π_M^{T*} are decreasing with λ , and This is because

consumers' willingness to pay and demand are both reduced by the uncertainty of value, so the manufacturer must reduce the wholesale price and the retailer must reduce the retail price as a way to attract consumers to buy the product. Interestingly, reduced consumer uncertainty about the value of the product would also cause the manufacturer to reduce the greenness of the product, which could be a result of the lack of consumer information about the product leading to mistrust of the greenness level of the product.

4.2. The blockchain-supported supply chain model

When the manufacturer chooses to apply blockchain technology, the consumer's uncertainty about the product information will be eliminated, so $\lambda = 0$. Similarly, the demand function is obtained:

$$D^B = 1 - \frac{p^B - g^B}{\theta} \quad (6)$$

The retailer's profit function:

$$\Pi_R^B = (p^B - w^B)D^B \quad (7)$$

The manufacturer's profit function:

$$\Pi_M^B = (w^B - c - b)D^B - \frac{1}{2}kg^{B2} \quad (8)$$

Lemma 2. In a blockchain-supported supply chain, By $\frac{\partial \Pi_B^B}{\partial p_B} = 0$, we have $p_B^*(w_B) = \frac{\theta + g_B + w_B}{2}$.

Substituting $p_B^*(w_B)$ into $\Pi_B^M(w_B, g_B)$, we have the following first-order conditions: $\frac{\partial \Pi_B^M}{\partial w_B} =$

$$\frac{\theta + g_B - 2w_B + b + c}{2\theta} = 0, \quad \frac{\partial \Pi_B^M}{\partial g_B} = \frac{-k\theta g_B + w_B - b - c}{\theta} = 0. \text{ Solving the above equations, we obtain } w_B^* =$$

$$\frac{2k\theta^2 + 2k(b+c)\theta - b - c}{4k\theta - 1}, \quad g_B^* = \frac{\theta - b - c}{4k\theta - 1}. \text{ We can deduce that the Hessian matrix is negative definite when } k > \frac{1}{4\theta}.$$

Therefore, when $k > \frac{1}{4\theta}$, Π_B^M is jointly concave in w_B^* and g_B^* , the equilibrium results are as

$$\text{follows: (1) Retail price, greenness level and wholesale price: } p^{B*} = \frac{3k\theta^2 + \theta(b+c)k - b - c}{4k\theta - 1}, \quad g^{B*} = \frac{\theta - b - c}{4k\theta - 1},$$

$$w^{B*} = \frac{2k\theta^2 + 2\theta(b+c)k - b - c}{4k\theta - 1}. \text{ (2) Demand: } D^{B*} = -\frac{k(-\theta + b + c)}{4k\theta - 1}. \text{ (3) Profits: } \Pi_R^{B*} = \frac{k^2\theta(b-\theta+c)^2}{(4k\theta-1)^2}, \quad \Pi_M^{B*} = \frac{k(b-\theta+c)^2}{8k\theta-2}.$$

Proposition 2. The equilibrium solution to the first order derivative of a has the following result:

$$\frac{\partial D^{B*}}{\partial b} < 0, \quad \frac{\partial g^{B*}}{\partial b} < 0, \quad \frac{\partial \Pi_M^{B*}}{\partial b} < 0, \quad \frac{\partial \Pi_R^{B*}}{\partial b} < 0, \quad \frac{\partial (w^{B*} - b - c)}{\partial b} < 0, \quad \frac{\partial (p^{B*} - w^{B*})}{\partial b} < 0.$$

Proposition 2 shows the first-order derivative, it is obvious to analyze that the demand for blockchain decreases as the operating cost increases. Although the monotonically decreasing cost of operating the blockchain cannot be derived from the first-order derivatives with respect to the retail and wholesale prices, it is evident from the manufacturer's margin ($w^{B*} - b - c$) and the retailer's margin ($p^{B*} - w^{B*}$) reduce that it decreases as the cost of operating the blockchain (b) increases. Thus, when

blockchain operating costs increase, both demand and profit margins decrease, and together they make manufacturers' and retailers' profits fall.

4.3. Impacts of adopting blockchain technology

While blockchain technology removes consumer uncertainty about product information, it also increases operating costs for manufacturers. Therefore, blockchain technology may not always be beneficial to manufacturers. In this subsection, we list the conditions under which manufacturers should adopt blockchain technology.

Lemma 3. By solving for $\Pi_B^{M*} - \Pi_T^{M*} = \frac{k^2\theta(b-\theta+c)^2}{(4k-1)^2} - \frac{k^2(c-1)^2}{(-\lambda^2+4k+2\lambda-1)^2} = 0$, When $b \leq b_1^*$, then $\Pi_B^{M*} \geq \Pi_T^{M*}$, the manufacturer should adopt supply blockchain technology if and only $b \leq b_1^*$, where

$$b_1^* = \theta - c + \frac{\sqrt{(-\lambda^2 + 4k + 2\lambda - 1)(4k\theta - 1)(c - 1)^2}}{(-\lambda^2 + 4k + 2\lambda - 1)}$$

Lemma 3 implies that if the operational cost of blockchain is in a certain range ($b \leq b_1^*$), the manufacturer should adopt blockchain technology. This is because the adoption or non-adoption of blockchain by the manufacturer has little impact on profits when the operating cost of blockchain is in a certain range, but at that point the benefits from blockchain do not exceed the operating cost of blockchain, so the technology should not be adopted at that point.

Proposition 3. When $b \leq b_1^*$, the manufacturers, the retailers and consumers can benefit from blockchain technology and thus increase social welfare.

Proposition 3 suggests that a manufacturer can increase its profit by using blockchain if the operating cost of blockchain is below a certain threshold. In addition, manufacturers will only tend to use blockchain if the costs of blockchain are small, and therefore retail prices will not increase significantly.

Comparing the optimal scenarios of using and not using blockchain technology, we arrive at the following proposition.

Proposition 4. When manufacturers are more willing to use blockchain technology,

(1) By solving for $p^{T*} - p^{B*} > 0$,

$$\text{when } c > \frac{-12\left(k - \frac{(\lambda-1)^2}{4}\right)k\theta^2 - 4\left((b-3)k - \frac{b(\lambda-1)^2}{4}\right)k\theta + (4b-3)k - b(\lambda-1)^2}{3(-1+(\lambda-1)^2\theta)k}, \quad p^{B*} < p^{T*}.$$

(2) By solving for $g^{B*} - g^{T*} > 0$, $w^{B*} - w^{T*} > 0$, $D^{B*} - D^{T*} > 0$, when $b <$

$$\frac{-12\left(k - \frac{(\lambda-1)^2}{4}\right)k\theta^2 - 4\left((b-3)k - \frac{b(\lambda-1)^2}{4}\right)k\theta + (4b-3)k - b(\lambda-1)^2}{3(-1+(\lambda-1)^2\theta)k}, \quad g^{B*} > g^{T*}, \quad w^{B*} > w^{T*}, \quad D^{B*} > D^{T*}.$$

(3) In the traditional supply chain model there are, $\frac{\partial p^{T*}}{\partial \lambda} < 0$, $\frac{\partial w^{T*}}{\partial \lambda} < 0$, $\frac{\partial g^{T*}}{\partial \lambda} < 0$, $\frac{\partial \Pi_T^{M*}}{\partial \lambda} < 0$.

(4) In the supply chain model there are, when $k > \frac{-b-c+3\theta}{4b\theta+4c\theta+4\theta^2}$ and $\theta > \frac{b+c}{3}$, $\frac{\partial p^{B*}}{\partial \theta} > 0$, $\frac{\partial g^{B*}}{\partial \theta} > 0$,

$$\frac{\partial w^{B*}}{\partial \theta} > 0, \quad \frac{\partial \Pi_B^{M*}}{\partial \theta} > 0.$$

Proposition 4-(1) indicates that adopting blockchain technology can lead to a reduction in retail prices when the manufacturer opts for blockchain usage, and the production cost surpasses a certain

threshold. This effect arises because blockchain technology ensures the product's value is fully appreciated by the consumer, which in turn elevates their expectations. Consequently, this might lead to an initial expectation that the retailer would set a higher retail price due to the increased consumer value perception.

Proposition 4-(2) demonstrates that if the value derived from blockchain falls below a specific threshold, it results in increased costs for the producer. Therefore, to compensate for these heightened costs, the producer is compelled to raise the wholesale price, necessitating a subsequent increase in the retail price by the retailer.

An intriguing observation from Proposition 4-(2) is that demand does not invariably rise with the adoption of blockchain technology. Specifically, blockchain adoption boosts demand when its operating costs are minimal, whereas high operating costs lead to a decrease in demand. This is attributed to the significant cost increases overshadowing the benefits of reduced uncertainty through blockchain, potentially diminishing consumer demand for the product.

Proposition 4-(3) shows that within a traditional supply chain, a decrease in consumer uncertainty leads to a reduction in the product's green level, retail price, wholesale price, and profit for the manufacturer. This trend might be explained by the limited impact of green products within the traditional model, which reinforces existing consumption patterns.

Finally, Proposition 4-(4) reveals that with the integration of blockchain technology, there is a positive correlation between retail price, wholesale price, product greenness, and the manufacturer's profit with consumer preference for blockchain technology, provided the cost coefficients and consumer preference for blockchain exceed a specific threshold.

5. Equilibrium analysis of green supply chain under government subsidy

To foster societal awareness of green supply chains and motivate enterprises to produce green products, governments typically offer certain subsidies to businesses. In this section, we delve further into the impact of government subsidies on green supply chains. Specifically, we investigate two types of government subsidies to enterprises: technology subsidies and output subsidies, respectively.

5.1. Modeling supply chains under technology subsidies

We use the superscript "I" to denote the case where the government provides a technology subsidy to the manufacturer, assuming that the technology subsidy coefficient is τ .

In a traditional supply chain, the manufacturer's profit function is:

$$\Pi_M^I = (w^T - c)D^T - \frac{1}{2}kg^{T^2}(1 - \tau) \quad (9)$$

In a supply chain using blockchain technology, the manufacturer's profit function is:

$$\Pi_M^{BI} = (w^B - c - b)D^B - \frac{1}{2}kg^{B^2}(1 - \tau) \quad (10)$$

Lemma 4. Consider government subsidies to manufacturers for technology, when $k > \frac{1}{4\theta(1-\tau)}$ (It

also holds true for the following)there exists an optimal manufacturer's wholesale price per unit of green product,the optimal product's greenness level, the optimal retail price, the optimal profit of each party involved, which maximizes social welfare and the profits of manufacturers and retailers. The proof is similar to that of Lemma 1 and Lemma 2.

(1) Under the traditional supply chain model. Retail price, greenness level and wholesale price:

$$p^{TI*} = \frac{(-1+\tau)(c+3)k+c(-1+\lambda)^2}{(4\tau-4)k+(-1+\lambda)^2}, g^{TI*} = \frac{\left((-1+\tau)(c+1)k+\frac{c(-1+\lambda)^2}{2}\right)(-1+\lambda)}{4k\left(k(-1+\tau)+\frac{(-1+\lambda)^2}{4}\right)(-1+\tau)}, w^{TI*} = \frac{2(-1+\tau)(c+1)k+c(-1+\lambda)^2}{(4\tau-4)k+(-1+\lambda)^2}, \Pi_R^{TI*} = \frac{(c-1)\left(\left((-1+\tau)^2k^2+2(-1+\lambda)^2(-1+\tau)k+\frac{(-1+\lambda)^4}{2}\right)c-(-1+\tau)^2k^2\right)}{16\left(k(-1+\tau)+\frac{(-1+\lambda)^2}{4}\right)^2}, \Pi_M^{TI*} = \frac{k(c-1)^2(-1+\tau)}{8k\tau+2\lambda^2-8k-4\lambda+2}.$$

(2) Under the supply chain of blockchain technology, $p^{BI*} = \frac{3\theta^2k(-1+\tau)+k(-1+\tau)(b+c)\theta+b+c}{1+4k(-1+\tau)\theta}, g^{BI*} = \frac{-\theta+b+c}{1+4k(-1+\tau)\theta}, w^{BI*} = \frac{2\theta^2k(-1+\tau)+2k(-1+\tau)(b+c)\theta+b+c}{1+4k(-1+\tau)\theta}, \Pi_R^{BI*} = \frac{(-1+\tau)^2(-\theta+b+c)^2k^2\theta}{16\left(\frac{1}{4}+k(-1+\tau)\theta\right)^2}, \Pi_M^{BI*} = \frac{(-\theta+b+c)^2(-1+\tau)k}{2+8k(-1+\tau)\theta}.$

Lemma 5. When governments subsidize manufacturers for technology, by solving for $\Pi_M^{BI*} - \Pi_M^{TI*} = 0$, When $b \leq b_2^*$, then $\Pi_M^{BI*} \geq \Pi_M^{TI*}$, the manufacturer should adopt supply blockchain technology if and only $b \leq b_1^*$, where

$$b_2^* = \theta - c + \frac{4\sqrt{\left(\frac{1}{4}+k(-1+\tau)\theta\right)(c-1)^2(k(-1+\tau)+\frac{(-1+\lambda)^2}{4})}}{\lambda^2 - 2\lambda + (4\tau - 4)k + 1}$$

Lemma 5 gives the conditions under which blockchain technology is beneficial to manufacturers if the government provides them with subsidies. This suggests that a manufacturer can increase its profit by using blockchain if the operating cost of blockchain is below a certain threshold. Note that the value of b_2^* depends on τ . This implies that the government subsidy policy influences the producer's decision regarding blockchain. The following proposition suggests further analysis.

Proposition 5. We can obtain by taking the first order derivative of b_2^* with respect to θ and τ :

$$\frac{\partial b_2^*}{\partial \theta} > 0, \frac{\partial b_2^*}{\partial \tau} > 0.$$

Proposition 5 reveals that b_2^* increases with respect to θ and τ . That is, when the degree of consumer preference for the application of blockchain technology θ or the value of the government's technology subsidy coefficient τ is larger, the larger the threshold for the conditions under which a manufacturer chooses to adopt blockchain technology, the more willing it is to adopt blockchain technology

Proposition 6. When manufacturers are more willing to use blockchain technology, by solving for $p^{BI*} - p^{TI*} > 0$, $g^{BI*} - g^{TI*} > 0$, $w^{BI*} - w^{TI*} > 0$, $D^{B*} - D^{T*} > 0$, by combining like conditions it follows that,

$$\text{when } \frac{4\tau k\lambda\theta - 4k\lambda\theta + \lambda^2\theta - 2\lambda\theta + \lambda + \theta - 1}{4\tau k + \lambda^2 - 4k - 2\lambda + 1} \leq b \leq -\frac{3k(-1+\tau)\left(-\frac{1}{4}+\left(k(-1+\tau)+\frac{(-1+\lambda)^2}{4}\right)\theta^2-\theta(-1+\tau)k\right)}{(1+\theta(-1+\tau)k)(k(-1+\tau)+\frac{(-1+\lambda)^2}{4})}, \text{ such}$$

$$\text{that } p^{BI*} < p^{TI*}, g^{BI*} > g^{TI*}, w^{BI*} > w^{TI*}, D^{B*} > D^{T*}.$$

Proposition 6 reveals that when the cost of operating the blockchain is in a certain range, the

adoption of blockchain technology decreases the retail price and increases the demand for the product. Specifically, the use of blockchain increases demand when the cost of operating the blockchain is in a certain range, while the adoption of blockchain leads to lower demand when the cost of operating the blockchain is high.

Proposition 7. When governments subsidize manufacturers for technology, the manufacturer is more likely to apply blockchain technology.

Solving for the first- and second-order derivatives of b_2^* with respect to τ , we find that b_2^* is increasing and convex in τ . Proposition 4 implies that public subsidy policies affect manufacturers' blockchain adoption strategies. This result is illustrated in more detail in Fig. 1, where "Strategy B" and "Strategy T" denote the strategy using blockchain and the strategy using traditional transactions, respectively. As shown in Fig. 1, when the government provides subsidies, manufacturers will use blockchain even if the cost of operating blockchain is relatively high. The reason for this is as follows.

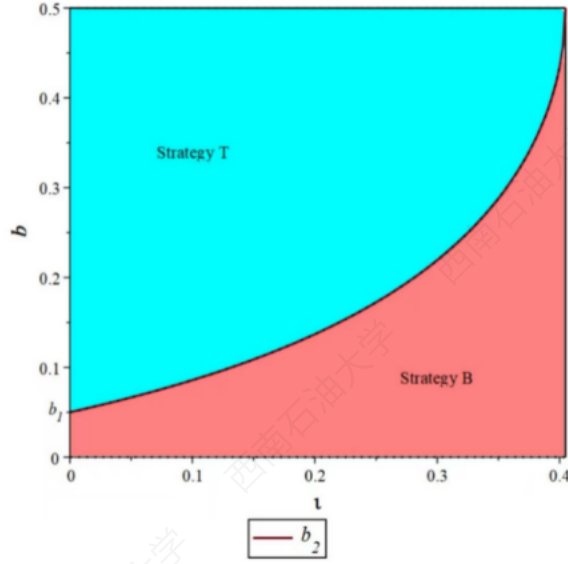


Fig. 1. Different strategies under Technology subsidy situation.

When the operating cost of blockchain exceeds a certain threshold, the producer's marginal profit is relatively low, whether or not the government provides a subsidy, but when the producer receives a subsidy from the government, demand increases. If the subsidy is large enough, the increase in the producer's operating costs will dominate the gains from eliminating consumer uncertainty and increasing demand. Thus, when the operating costs of blockchain are relatively high, the producer remains profitable and the probability of using blockchain increases. In our model, the producer subsidy system is equivalent to allocating part of the producer's ecological costs. Thus, companies with the advantage of low ecological production costs may be more likely to adopt blockchain in practice.

5.2. Modeling Supply Chains under Production Subsidies

We use the superscript "J" to denote the case where the government subsidizes the production of a manufacturer, assuming a subsidy per unit of green product μ .

In a traditional supply chain, the manufacturer's profit function is::

$$\Pi_M^{TJ} = (w^T - c + \mu)D^T - \frac{1}{2}kg^{T2} \quad (11)$$

In a supply chain using blockchain technology, the manufacturer's profit function is:

$$\Pi_M^{BJ} = (w^B - c - b + \mu)D^B - \frac{1}{2}kg^{B2} \quad (12)$$

Lemma 6. Considering the government's output subsidy to the producer, there exists an optimal wholesale price of the producer's green product per unit, an optimal degree of greenness of the product, an optimal retail price, and an optimal profit for all parties involved, which maximizes the social welfare and the profits of the producer and the retailer. The proof is similar to those of Lemma 1 and Lemma 2.

(1) Under the traditional supply chain model. Retail price, greenness level and wholesale price:

$$p^{TJ*} = \frac{(-c+\mu)\lambda^2+(2c-2\mu)\lambda+(-k+1)\mu+(k-1)c+3k}{-\lambda^2+4k+2\lambda-1}, g^{TJ*} = \frac{(c-\mu-1)(-1+\lambda)}{-\lambda^2+4k+2\lambda-1},$$

$$w^{TJ*} = \frac{(-c+\mu)\lambda^2+(2c-2\mu)\lambda+(-2k+1)\mu+(2k-1)c+2k}{-\lambda^2+4k+2\lambda-1}, \Pi_R^{TJ*} = \frac{k^2(c-\mu-1)^2}{(-\lambda^2+4k+2\lambda-1)^2}, \Pi_M^{TJ*} = \frac{k(c-\mu-1)^2}{-2\lambda^2+8k+4\lambda-2}.$$

(2) Under the supply chain of blockchain technology, $p^{BJ*} = \frac{3\theta^2k+k(b+c-\mu)\theta-b-c+\mu}{4\theta k-1}$, $g^{BJ*} = \frac{-b-c+\mu+\theta}{4\theta k-1}$, $w^{BJ*} = \frac{2\theta^2k+2k(b+c-\mu)\theta-b-c+\mu}{4\theta k-1}$, $\Pi_R^{BJ*} = \frac{\theta k^2(b+c-\mu-\theta)^2}{(4\theta k-1)^2}$, $\Pi_M^{BJ*} = \frac{k(b+c-\mu-\theta)^2}{8\theta k-2}$.

Lemma 7. When the government subsidizes manufacturers' output, the proof is similar to those of Lemma 3, the manufacturer only introduces blockchain technology if $b \leq b_3^*$, where

$$b_3^* = \theta - c + \mu + \frac{c - \mu - \theta - \sqrt{(4\theta k - 1)(-\lambda^2 + 4k + 2\lambda - 1)(c - \mu - 1)^2}}{-\lambda^2 + 4k + 2\lambda - 1}$$

Lemma 7 gives a threshold condition for the use of blockchain when the government subsidizes retailers. We observe that b_3^* depends not only on λ , but also on θ, μ and c . Note that b_3^* not necessarily monotonic in μ .

Proposition 8. When the government subsidizes manufacturers' output, the manufacturer is more likely to apply blockchain technology.

By solving for the first and second order derivatives with respect to μ , we find that b_3^* exhibits increasing and convex properties with respect to μ . Proposition 8 implies that the government subsidy policy has a significant impact on the blockchain adoption strategy of manufacturers. This result is illustrated in more detail in Fig. 2, where "Strategy B" and "Strategy T" represent the strategies of blockchain adoption and traditional operation, respectively. Observing Fig. 2, it can be seen that even in the case of government subsidies, manufacturers still tend to adopt blockchain when the operating cost of blockchain is relatively high. The reason for this phenomenon is that when the operating cost of blockchain exceeds a certain threshold, the manufacturer's marginal profit is relatively small, regardless of whether the government provides subsidies or not. However, demand tends to rise in cases where manufacturers receive government subsidies. With a sufficiently large subsidy, the increase in the manufacturer's operating costs will be compensated by the reduction in consumer uncertainty and the increase in demand. As a result, even with relatively high operating costs of blockchain, manufacturers are still able to realize substantial profits, thereby increasing the likelihood of blockchain adoption. In our model, the manufacturer's subsidy program is equivalent to sharing a portion of the green costs. As a result, firms that have a low-cost advantage in green production are more likely to choose to adopt blockchain in practice.

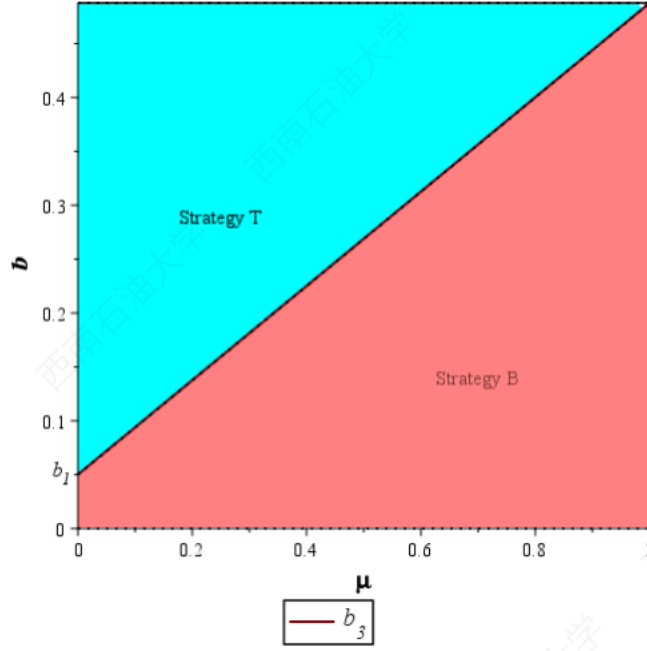


Fig. 2 Different strategies under Output subsidy situation.

Proposition 9. As shown in Fig. 3, when the government subsidizes manufacturers' output, manufacturers' profits rise with μ .

Proposition 9 shows that if the government subsidizes producers, it will have a positive impact on manufacturers. When the production subsidy coefficient is at less than the threshold, manufacturers do not use blockchain technology because of profits. When the subsidy coefficient is above the threshold, manufacturers will actively adopt blockchain technology. This result is important for the government because it provides a useful theoretical framework for supporting firms to develop blockchain technology.

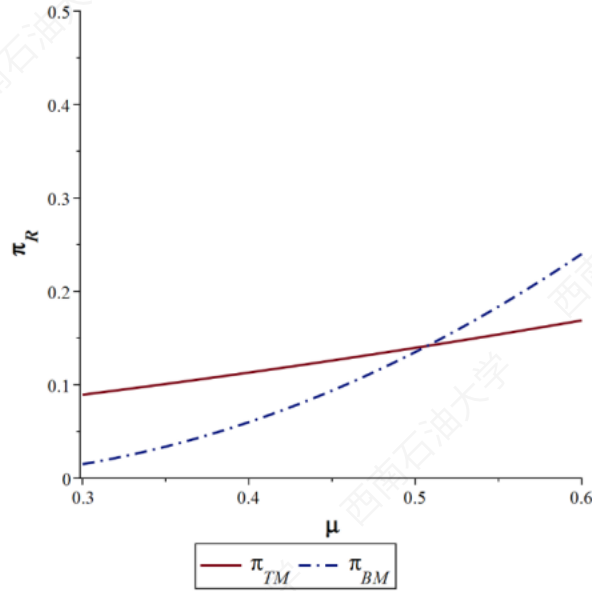


Fig. 3 Effect of production subsidy amount μ on profit

Proposition 10. As shown in Fig. 1 and Fig. 2, when the government subsidizes manufacturers for

technology, do manufacturers have a greater probability of preferring a traditional supply chain. When governments subsidize manufacturers' output, manufacturers have a greater probability of preferring supply chains that use blockchain technology.

Proposition 10 highlights the influence of various subsidy models on the adoption of blockchain technology. The rationale behind Proposition 10 can be explained as follows: Manufacturers might not see an increase in profits from adopting blockchain technology if the operational costs associated with blockchain are relatively high. This insight suggests to the government that if it aims to encourage manufacturers to utilize blockchain for industrial upgrading, it would be beneficial for the government to provide subsidies for the manufacturers' output.

6. Conclusions

We examine a green supply chain system comprising a single manufacturer and a single retailer, analyzing the optimal decision-making within the green supply chain without government subsidies and determining the conditions under which the manufacturer would adopt blockchain. This analysis is conducted by constructing a Stackelberg model, where the manufacturer is the leader and the retailer is the follower. Subsequently, we explore the optimal decision-making for the green supply chain under two scenarios of government subsidies to the manufacturer: technology subsidies and production subsidies, among other considerations. The study yields the following conclusions:

Firstly, in the scenario without government subsidies: (1) Within the traditional supply chain, the level of product greenness, retail price, wholesale price, retailer's profit, and manufacturer's profit all decline as consumer uncertainty decreases. (2) In a blockchain-enabled supply chain, market demand diminishes as the operating costs of the blockchain increase, meaning manufacturers only adopt blockchain technology when its operating costs are low. Although the operating costs of blockchain influence retailers' decisions, the traceability mechanism offered by blockchain, which mitigates consumer uncertainty to a degree, benefits the retailers' profits.

Secondly, in the scenario with government subsidies: (1) When the government provides technology subsidies to manufacturers, they will choose to adopt blockchain technology only if the operating cost of blockchain is below a certain threshold. In this scenario, manufacturers can achieve higher profits than in the traditional supply chain. Furthermore, this threshold value increases in relation to the degree of consumer preference for blockchain technology and the coefficient of the government's technology subsidy. That is, the greater the consumer preference for blockchain technology or the larger the government's technology subsidy coefficient, the higher the threshold for adopting blockchain technology becomes, encouraging more manufacturers to consider its adoption. (2) Similarly, when the government subsidizes the manufacturer's output, there exists a threshold below which the cost of operating the blockchain encourages the manufacturer to adopt the technology. This threshold is influenced by the degree of consumer preference for blockchain technology, the government's technology subsidy coefficient, consumer uncertainty, and other factors. As the government's output subsidy coefficient increases, manufacturers are more inclined to adopt blockchain technology; the larger the government's output subsidy coefficient, the more attractive blockchain technology adoption becomes to manufacturers. These findings indicate that enterprises should weigh these factors collectively when deciding on the adoption of blockchain technology.

Finally, our study opens several avenues for future research. We derived the demand function from the consumer's utility function and then constructed the profit function from it to analyze the optimal

decision-making within green supply chains under the influence of government subsidies and blockchain technology, as well as to determine the conditions under which manufacturers would adopt blockchain. Future studies could position the government as a decision-maker, comparing various government-led subsidy policies to explore their impacts on efficiency and other outcome variables.

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